

Thermal Queue Governor

Patent review and virtual-motor validation preparation

Document authors: Srinivas Nampalli and Saathvik Gampa. September 9, 2026. AI-assisted technical review; this supplement does not replace the preserved original study or dossier.

The owner reports equal (50/50) general technical contributions. Specific contributions, dates, supporting conception records, and patent inventorship remain unresolved. The project maintainer is SrinivasNampalli. Original-material rights remain reserved pending review.

Review outcome

The expanded review maps all eleven discussion claims to eleven primary-source groups and records twenty-five exact search queries. There is substantial prior-art overlap in saturation-aware estimation, delayed command governors, thermal current limits, pending-command scheduling, finite-prefix safety checks, safe continuation, and quantized input selection. The evidence does not establish that this invention is unique.

The exact interaction of a censor-consistent scalar interval, an immutable accepted FIFO, every-prefix feasibility, maximum admissible grid effort, and a zero-input tail remains a narrow research subject. No inspected primary document was identified as disclosing every element together; the limited search does not establish its absence from the wider literature or make a predictable combination inventive. The detailed claim matrix and comparison locations follow in this packet.

Completed review fields

Field	Recorded state
Working project title	Thermal Queue Governor
Technical subject	Conditional scalar thermal command admission with clipped sensing and accepted FIFO commands
Document authors	Srinivas Nampalli; Saathvik Gampa
Contribution statement	User-reported equal general technical contribution; detailed allocation and dates not supplied
Implemented embodiment	Existing synchronous Python governor and interactive scalar prototype
Claims reviewed	Existing eleven discussion claims; original text preserved
Broad novelty assessment	Strong documented overlap; broad unique-invention framing unsupported
Narrow patentability	Unresolved; requires further combination analysis and professional review
Patent inventors, applicant, ownership	Not established by document author names or the 50/50 statement
Filing jurisdiction	Not selected; prior US-style discussion drafting is not a filing decision
Filing/submission status	No application, declaration, or inventor signature submitted by this work
Public disclosure	GitHub publication recorded separately; complete earliest-disclosure facts remain unresolved
Selected motor	Owner will provide the exact model and specifications
New motor-specific calibration / held-out episodes	Zero / zero
Physical motor test episodes	Zero

What the new test harness represents

The independent illustrative plant models drive current, winding temperature, housing temperature, and sensor temperature. Current-squared heating depends on temperature-dependent copper resistance. Separate thermal masses and conductances represent stored heat and cooling. A lagged sensor is clipped after bounded noise. The actual simulated winding state is never clipped.

The code separates requested effort, accepted FIFO effort, reported applied command, and actual simulated RMS current. Current can continue producing heat after a zero command. This is a useful challenge to the original model's actuator assumptions. It is an averaged thermal/current model, not a full electrical circuit, electromagnetic, speed, torque, or mechanical-load simulator.

The harness includes synthetic numerical and integration checks. Such checks verify software equations and accounting against artificial fixtures; they do not calibrate or validate the motor the owner intends to select. No generic study was run and relabeled as that motor. The draft protocol blocks study execution until it is finalized for the selected motor.

Validation sequence and decision gates

1. Obtain the manufacturer, exact motor/variant, datasheet, and intended drive/controller.
2. Assess model suitability and record parameter sources, units, mounting/cooling conditions, sensor placement, and real operating limits. Do not import the illustrative 105 C threshold as a product specification.
3. Publish the finalized protocol and source/configuration identity before data collection. Keep calibration and held-out seeds/conditions disjoint.
4. Estimate declared allowances using calibration data only. A finite empirical envelope is not a global physical bound. Preserve model infeasibility and failed checks rather than tuning them away after testing.
5. Run paired held-out comparisons with clipping, dropout, changing ambient/load, and falsely reported actuation. Report winding peaks, containment misses, operating and reference-limit breaches, command throughput, and command/current differences separately.
6. Compare numerical resolutions and retain the raw traces and computing-environment receipt. A run on a GitHub-hosted VM verifies software execution, not a physical motor.
7. Plan independent physical measurements and protective hardware before any bench validation. No physical experiment has occurred in this phase.

Evidence boundaries and next inputs

The historical study remains 1,120 final/post-main episodes and 1,008,000 transitions, with a separate 900-episode pilot. Its 90.4% effort and zero sampled breaches refer to 120 primary TQG episodes satisfying that scalar model's assumptions. They are not measurements from the new multi-state plant or a real motor. The original 47 Python tests passed again during this review, and the primary evidence recalculation reproduced the preserved figures without adding episodes.

The immediate required input is the motor's exact model and datasheet, plus its drive/controller if known. Human contribution records need actual technical ideas and approximate dates; an invented split would not provide that evidence. Filing decisions also require the missing jurisdiction, ownership, and disclosure facts.

Current records are linked from the [GitHub repository](#): the [search log](#), [process log](#), [contribution record](#), and [draft harness protocol](#). The packet's following sections preserve the detailed review and motor intake worksheet.

Thermal Queue Governor: focused claim-level technical review

Review date: **9 September 2026**. This is a bounded technical prior-art review of the eleven existing discussion claims, not a patentability, freedom-to-operate, or filing opinion. No application was filed and no third party was contacted. The search and access log records the queries, documents inspected, access failures, and remaining coverage limits. The earlier `prior_art_challenge.md` is preserved as a historical review.

Decision

The evidence does not support calling TQG unique or establishing a patentable invention. Its broad ingredients have substantial documented overlap. The useful narrow discussion is the exact interaction of one-sided censor-consistent state bounds, an immutable accepted FIFO, every-prefix feasibility, a greatest admissible grid effort, and a verified zero-input tail in a positive scalar model. No inspected primary document was identified as disclosing that entire exact combination together. That limited search result is not a novelty guarantee, and the combination challenge remains strong.

This review adds particularly relevant challenges: an older motor patent derives current limits from predicted thermal power, including a square-root operation; another uses voice-coil temperature and cooling delay when selecting pending commands. Their specific disclosures make “thermal budget,” “square-root current cap,” and “temperature-aware command queue” weak standalone distinctions. These are technical comparisons, not assertions that their complete claims cover TQG. See P1 and P2 below.

Exact subject reviewed

The implemented `interval_queue` policy in `governor.py` maintains a finite interval for nonnegative temperature rise, uses $x_{\text{next}} = a \cdot x + b \cdot u^2 + w$, intersects it with the observation-consistent set, predicts all already accepted FIFO actions, and appends one admissible quantized action. Its model constructor checks $w_{\text{max}} / (1 - a_{\text{max}}) \leq R$. `advance` uses a caller-asserted applied action. Invalidity is latched locally.

The equations discussed here describe ideal arithmetic. The implementation uses $1e-9$ comparison tolerances and a $1e-12$ grid-floor adjustment; it does not use directed rounding. Neither this review nor the existing simulation results establishes a strict machine-arithmetic enclosure theorem.

This is a synchronous software contract. The code does **not** authenticate acknowledgments, verify physical current, enforce wall-clock command expiry, reject network replay, calibrate a motor, or establish winding/hotspot or intersample safety. The diagnostic decision counter is not an execution permit. A virtual motor experiment remains a simulation; it cannot become physical motor validation by running on a VM or adding hardware graphics.

Reviewed file snapshots, SHA-256:

File	SHA-256
<code>project/docs/CLAIMS_DISCUSSION_DRAFT.md</code>	25df0b2a34f0a0ccb60917b740aef59295b6776d41b947275d8dd575c989a906

File	SHA-256
project/src/governor.py	9cccc5f82ad0aaa759e379471874bd7fa163d0580c920f4d033ae14a07b6975f

The checkout HEAD observed during review was f4d438f10de18979f47e237457942ce143679abf; hashes identify the actual reviewed files independently of later publication edits. This analysis neither assigns human inventorship nor infers individual contributions from an author name.

Primary references and precise comparison points

The identifiers below are local to this review; they do not replace the earlier review's numbering. “Not identified” means not located in the inspected portions, not proven absent from every version, family member, or other reference.

L1

Primary source, date, and inspected location: Yang and Li, “Set-membership filtering for systems with sensor saturation”, *Automatica* 45, 1896-1902; online **27 May 2009**. [Author PDF](#), §2 eqs. (1)-(9), §3 Theorem 1, Remarks 1-2.

Supported overlap and distinction: Saturation-aware bounded-set estimation and a bounded initial set are explicit. Its equation adds noise **after symmetric saturation**; TQG's claim 2 puts bounded noise **before one-sided clipping**. Its ellipsoidal/sector construction is not TQG's exact scalar half-line intersection. Remark 1 limits its stated study to occasional saturation. These differences should remain visible.

L2

Primary source, date, and inspected location: Shen et al., “Constrained Control for Cloud Robotic Under Time Delay Based on Command Governor With Interval Estimation”, *IEEE Access* 7, 70999-71006, **2019**, DOI 10.1109/ACCESS.2019.2920017. [Original paper text mirrored on ResearchGate](#), §§III-IV, eqs. (19)-(21), (31)-(32).

Supported overlap and distinction: Stores command history, bounds delayed state, and modifies a request using interval-constrained prediction. Its uncertain communication delay/reference-governor formulation differs from an exact accepted FIFO and a scalar quadratic thermal model. A noise-before-clipping update was not identified. Publisher access failed; see the version/date caveat in the log.

L3

Primary source, date, and inspected location: Rosenthal, Noack and Hanebeck, “State Estimation in Networked Control Systems With Delayed and Lossy Acknowledgments”, *2017 MFI*, pp. 435-441, DOI 10.1109/MFI.2017.8170359. [Author PDF](#), §§II-III; also the [extended author chapter](#), §2 Remarks 1-2 and §3.1.

Supported overlap and distinction: Tracks still-applicable input history and models delayed/lost actuator acknowledgments. The buffer replaces old sequences with the newest time-indexed sequence; it is not TQG's immutable FIFO. Its Gaussian/MMSE estimator is not a deterministic thermal interval. The extended chapter distinguishes transport receipt from actuator acknowledgment and describes zero-input fallback.

L4

Primary source, date, and inspected location: Wabersich and Zeilinger, “A predictive safety filter for learning-based control of constrained nonlinear dynamical systems”, [arXiv history](#): v1 **13 December 2018**, inspected v4 **17 May 2021**. [v4 PDF](#), §4.1 eq. (5), Algorithm 1; §4.2 eq. (6), Algorithm 2.

Supported overlap and distinction: Admits or modifies requested input using a constrained trajectory and terminal safe set, with backup continuation. No exact clipped-temperature FIFO specialization was identified. Its uncertain-model formulation uses probabilistic assurance; do not substitute it for TQG's deterministic scalar assumptions.

L5

Primary source, date, and inspected location: Bemporad and Garulli, *Predictive Control via Set-membership State Estimation for Constrained Linear Systems with Disturbances*, **July 1997** ECC, identified in the [author bibliography](#). [Author PDF](#), §2 measurement intersection/prediction, eqs. (7), (9)-(12); §3 Theorem 1, eqs. (13)-(14).

Supported overlap and distinction: Already combines set-valued estimation with robust future constraint enforcement. Theorem 1 reduces infinitely many constraints to a finite check plus an additional tail condition. Its linear, unsaturated-measurement, optimized sequence/constant-tail setting does not expressly disclose TQG's particular accepted queue, one candidate, and zero-tail calculation.

L6

Primary source, date, and inspected location: Bemporad, *Predictive Control of Teleoperated Constrained Systems with Unbounded Communication Delays*, **1998** CDC, pp. 2133-2138, DOI 10.1109/CDC.1998.758651. [Author PDF](#), §2, eqs. (1)-(4), Fig. 1 and synchronization logic.

Supported overlap and distinction: Buffered predictive commands can provide emergency continuation through communication delay. This was reached through L3's reference 5. It challenges a broad "buffer plus safe fallback" proposal, not the complete present scalar claim.

L7

Primary source, date, and inspected location: Danielson and Di Cairano, *Robust Soft-Landing Control with Quantized Input*, **August 2016**, IFAC-PapersOnLine 49(18), 35-40, DOI 10.1016/j.ifacol.2016.10.136. [MERL record](#), [author PDF](#), §§3.2-3.3, eqs. (7), (11).

Supported overlap and distinction: Combines constrained predictive control, rounding into a finite input alphabet, and robust invariance. Its nearest-input rounding with modeled rounding error differs from TQG's monotone downward scalar grid floor. Neither quantization nor constraint-preserving rounding is new at the general level.

P1

Primary source, date, and inspected location: **US7248009B1**, *Motor temperature control using estimated motor temperature based on motor power dissipation*, published **24 July 2007**. [Patent record and original text](#), [issued PDF](#), claims 1, 6-11; Figs. 4-5, eq. (3).

Supported overlap and distinction: Motor thermal prediction, current-derived dissipation, future maximum permitted dissipation, and PWM limiting are explicit. Fig. 5 uses I^2R and a square-root current limit. No interval observer, clipped-sensor half-line, or immutable FIFO every-prefix test was identified. The square-root operation alone is particularly weak as a distinction.

P2

Primary source, date, and inspected location: **US7114029B1**, *Disk drive employing a multi-phase rotational position optimization (RPO) algorithm*, published **26 September 2006**. [Patent record](#), [issued PDF](#), claims 1/4 and 6/9; Figs. 5A-5B.

Supported overlap and distinction: Selects among pending disk commands using a voice-coil motor temperature parameter. The description converts temperature to cooling delay before another seek. This is motor-related thermal queue scheduling, but it reorders/selects commands rather than certifying an immutable accepted FIFO under censored sensing.

P3

Primary source, date, and inspected location: **US8773058B2**, *Rotor temperature estimation and motor control torque limiting for vector-controlled AC induction motors*, published **8 July 2014**. [Original patent text](#), claims 1-6, Fig. 5.

Supported overlap and distinction: Runtime multi-node thermal estimation uses stator temperature/current to limit motor power. This supports the broad thermal-estimation/derating overlap; it does not establish the claimed clipping/queue combination.

P4

Primary source, date, and inspected location: **US11994926B2**, *Drive performance, power, and temperature management*, published **28 May 2024**. [Original patent text mirrored by Justia](#), claims 1, 5-7; description Figs. 5 and 10.

Supported overlap and distinction: Dependent claims 5-7 use pending workload and command-queue details for future power estimates. The description also predicts temperature and limits performance. The independent claim includes specific ADC/power-rail measurement features. Do not describe the thermal narrative as the entire issued claim or equate it with TQG's exact equations.

Dates above identify the inspected publications, not verified earliest entitlement dates or current enforceability. Original patent text/PDF is evidence of what a publication says, even when hosted by a mirror. Legal status and complete family/prosecution histories were not audited.

Eleven-claim overlap matrix

S means supported overlap in the cited location. **N** means the exact narrower limitation was not identified there. **U** means its distinctiveness or complete coverage remains unresolved. None of these labels is a legal determination that a whole claim is anticipated, valid, or infringed.

1, full nine-part method

S: supported overlap / nearest challenge: L1 estimation; L2 delayed governor; L3 input history; L4/L5 constrained prefix and tail; P1/P3 motor thermal limiting.

N: exact feature not identified in those inspected passages: The complete combination of the specified positive scalar recurrence, clipping intersection, exact FIFO, one grid candidate, invariant zero tail, and persistent latch was not found together in one inspected document.

U: assessment of the remaining difference: High combination risk. A narrower implementation interaction is a discussion candidate, not established uniqueness. Every clause must be compared as a combination, not just the title.

2, noise-before-clipping and exact half-line

S: supported overlap / nearest challenge: L1 §2 gives an explicit saturation/noise model; L5 §2 uses measurement-consistent sets.

N: exact feature not identified in those inspected passages: L1 eq. (2) places noise after symmetric saturation, unlike claim 2.

U: assessment of the remaining difference: The sensor ordering is a real distinction. Once this observation law is chosen, the half-line is its direct algebraic inverse; that alone supplies no demonstrated inventive effect.

3, prediction-only missing reading; retain model upper bound while clipped

S: supported overlap / nearest challenge: L1 bounded-set estimation; L5 §2 set intersection and prediction.

N: exact feature not identified in those inspected passages: The literal None branch and this precise scalar upper-bound rule were not identified.

U: assessment of the remaining difference: Consistency with unavailable/censored information explains the implementation. A software branch or the avoidance of an incorrect exact-clip update does not itself establish novelty.

4, lower/upper corner recurrence using asserted applied effort

S: supported overlap / nearest challenge: L5 §2 propagates a compatible state set with known input and bounded disturbance; L3 incorporates applied-input information.

N: exact feature not identified in those inspected passages: The exact $a_{\min}/\max$, $b_{\min}/\max \cdot u^2$, $w_{\min}/\max$ scalar formulas were not located verbatim.

U: assessment of the remaining difference: Endpoint arithmetic follows the selected nonnegative monotone model. The general enclosure principle is already supported by L5.

5, complete accepted FIFO; every-prefix maximum; zero does not cancel prior commands

S: supported overlap / nearest challenge: L2 §§III-IV command-history prediction; L3 §§II-III applicable inputs; L5 eq. (9) all-future constraints; P2 thermal pending-command control.

N: exact feature not identified in those inspected passages: L3 replaces sequences; P2 selects/reorders commands. Neither is this irrevocable accepted FIFO plus every-prefix maximum.

U: assessment of the remaining difference: This is a meaningful actuator-contract distinction and the best place to sharpen the implementation argument. Its inventiveness remains unresolved against constrained prediction with fixed input prefixes.

6, square-root cap, downward grid floor, final recomputation

S: supported overlap / nearest challenge: P1 Fig. 5 square-root current limiting; L7 §3.2 quantized constrained input selection.

N: exact feature not identified in those inspected passages: The exact post-FIFO scalar formula and one-sided grid floor with this final check were not identified together.

U: assessment of the remaining difference: These steps solve one monotone quadratic inequality on a uniform grid. Their usefulness is supported by the code; an unexpected technical effect over the closest combination is not established.

7, $w_{\max}/(1-a_{\max}) \leq R$ gives an indefinite zero tail

S: supported overlap / nearest challenge: L4 §4.1 terminal set; L5 Theorem 1 finite check/tail condition; L3 §II zero-input strategy.

N: exact feature not identified in those inspected passages: This specific scalar equilibrium inequality was not located in those passages.

U: assessment of the remaining difference: For the selected recurrence it follows from $a_{\max} * R + w_{\max} \leq R$. This is a model-specific invariant test, not evidence of a new general safety principle.

8, reject unbounded commissioning or non-invariant model

S: supported overlap / nearest challenge: L1 §2 bounded initial set; L5 Assumption 3 feasible initialization; P1 claims 3-4 initialize a thermal estimate.

N: exact feature not identified in those inspected passages: The exact constructor rejection rules were not located.

U: assessment of the remaining difference: Validating assumptions improves implementation reliability but cannot confirm containment of unknown physical temperature. No distinctive technical effect is established by malformed-input rejection alone.

9, 120-entry/finite/range checks before reading

S: supported overlap / nearest challenge: L3 uses finite histories, but not this precise queue-validation ordering.

N: exact feature not identified in those inspected passages: The literal 120-entry maximum and exact validation precedence were not identified.

U: assessment of the remaining difference: Arbitrary capacity and ordering can narrow text without demonstrating an inventive interaction. The reason for the threshold would need engineering support.

10, persistent invalid latch without convenient rebase

S: supported overlap / nearest challenge: L4 has a different backup-continuation policy; it is not the same local latch.

N: exact feature not identified in those inspected passages: No cited passage was found to disclose this exact instance-lifetime latch and no-rebase behavior.

U: assessment of the remaining difference: **Unresolved, not a proven novelty point.** The present evidence does not compare this rule with a separately validated recommissioning scheme or establish a non-routine benefit.

11, invalid asserted applied value latches and prevents propagation

S: supported overlap / nearest challenge: L3 makes applied-input uncertainty explicit.

N: exact feature not identified in those inspected passages: The exact scalar range-check/latch branch was not located.

U: assessment of the remaining difference: Range validation is not authentication or evidence of execution. TQG assumes an authoritative scalar rather than solving L3's harder uncertainty problem. That simpler interface is not an established inventive contribution.

What might be worth developing

The most plausible narrow research subject is **preserving the relationship between an irreversible accepted heat-producing prefix and a censor-consistent bound at the moment of admission**. The source code already expresses that relationship in a small, inspectable scalar calculation. However, none of the current empirical results establishes that this combination gives an unexpected result over the strongest properly adapted predecessor. Breaking an observer on purpose or omitting queued inputs provides mechanism evidence, not proof that all reasonable prior controllers fail.

Two distinctions should be tested without inflating them into delivered functionality:

1. **Present implementation, stronger evidence needed:** compare the exact finite-prefix/grid/zero-tail procedure against a correctly formulated constrained predictor with the same sensing, uncertainty, and immutable queue assumptions. Match action grid, admissible information, fallback assumptions, and computation budget. Report equivalence if the decisions coincide; do not tune the competitor into failure. Examine extended clipping and cases where a queued prefix already makes recovery impossible.

2. **Future proposal, not implemented by the eleven claims:** maintain a sound set of possible applied-action histories when acknowledgments or timing are uncertain, then certify admission against every consistent history and a physically justified fallback. L3 and L6 already make history/network uncertainty and buffered continuation substantial prior-art obstacles. Merely adding timestamps, tokens, expiry, or acknowledgments is not enough. A distinct technical result and new support would have to be demonstrated before claiming such an extension.

For virtual motor validation, a more detailed plant can reveal whether the scalar observer's assumptions survive sensor lag, multiple thermal nodes, cooling changes, or mismatched actuation. It cannot establish real parameter bounds, sensor placement, physical protection, or legal uniqueness. Any extension must retain negative cases, avoid feeding hidden simulated truth to the controller, and label post-review experiments separately from the frozen original study.

Confidence and remaining uncertainty

This review supports **high confidence in broad overlap** and **low confidence in any patentability conclusion for the narrow combination**. It neither establishes that an exact full disclosure is absent nor concludes that every narrower claim necessarily fails. A claim-focused professional review should now examine the exact combination, supported human contributions, public-disclosure chronology, and relevant jurisdiction using the actual cited documents.

The search did not complete classification/citation closure, non-English full-text review, all patent families, prosecution histories, or current legal-status checks. Two recent secondary-only leads could not be verified from a primary publication and are quarantined in the log. They are not used to claim an exact match or a confirmed date. The supplied paper's authorship/publication history and the repository's actual public-disclosure dates are separate factual questions; an author credit is not a determination of patent inventorship.

Motor specification intake

Status on September 9, 2026: **awaiting the motor model and specifications from the owner.** The owner selected a specific motor instead of proceeding with a generic validation study. No particular product has been inferred or selected on the owner's behalf.

First information needed

Provide the manufacturer and exact model/part number, preferably with a datasheet link or file. Identify the intended drive/controller if known. This is enough to begin checking what the manufacturer documents and what still needs measurement.

Parameter and evidence worksheet

Item	Current value or status	How it will be established
Manufacturer, motor family, exact part number	Not provided	Owner's selection and manufacturer datasheet
Motor topology and winding connection	Not provided	Manufacturer data; affects how current and resistance map to copper losses
Rated supply, current, speed, and load	Not provided	Datasheet plus intended operating point
Winding resistance and reference temperature	Not provided	Datasheet and, when available, calibrated measurement; distinguish phase from line-to-line values
Winding and housing thermal properties	Not provided	Manufacturer thermal data, mounting conditions, and identification experiments
Continuous and transient temperature limits	Not provided	Manufacturer specifications and application derating
Sensor type, position, accuracy, lag, and clipping ceiling	Not provided	Sensor/controller specifications and calibration
Driver, current regulation, current limit, PWM, and shutdown behavior	Not provided	Driver documentation and measurements
Cooling, enclosure, mounting, ambient range, and duty cycle	Not provided	Intended setup, photographs/drawings, and measured boundary conditions
FIFO latency, command units, timestamps, and actual-current feedback	Not provided	Controller/communications documentation and logged execution evidence
Independent reference temperature and protective cutoff	Not provided	Instrument selection and a bench-test procedure before physical testing

A datasheet value is not automatically a verified worst-case bound. Unknown parameters remain unknown; illustrative defaults must not silently become specifications for the selected motor. A winding-temperature estimate and a casing-mounted sensor reading are distinct quantities.

Validation gates

1. Record the exact motor, source documents, units, and applicable operating conditions.
2. Check whether the available averaged current/two-node thermal model can represent that motor and drive. Extend or replace it when required; for example, its present form does not model speed, back EMF, torque, or individual phase waveforms.
3. Establish parameter ranges and a commissioning procedure from documented or measured evidence. Reserve separate calibration and held-out conditions.
4. Publish a versioned protocol and source/configuration hashes before calibration and held-out evaluation. Select the real temperature limits, rather than importing the generic 105 C example.
5. Run software-in-the-loop trials in the documented computing environment. Keep synthetic truth separate from sensor/controller inputs and report all failures.
6. Assess the model against independent physical measurements when equipment and a test procedure are available. A VM run is software execution and cannot replace this gate.

The existing 1,120-episode scalar study and its headline performance numbers remain historical synthetic results. There are currently **zero motor-specific validation episodes and zero physical-motor test episodes** in this new validation phase.